

Binomial Expansions

Worked Solutions with TikZ Diagrams

Wisest Maths — Year 1 A-Level, Binomial Expansions (ref **be**)

This document contains all 85 *Binomial Expansions* questions, each with a fully worked solution and a TikZ diagram matched to the method:

- **Pascal's triangle** (row n highlighted) for expanding $(a + b)^n$;
- the **general-term card** $T_{r+1} = \binom{n}{r} a^{n-r} b^r$ for finding a coefficient or the term independent of x ;
- a **factorial card** for evaluating $\binom{n}{r}$;
- **solution-flow** diagrams for approximations, products, finding unknowns, and identities.

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Binomial Expansion 01 (be-001)

Difficulty: Foundation / Marks: 3

Expand $(1+x)^5$ in ascending powers of x , giving all terms.

Worked solution.

Step 1: Apply the binomial theorem: $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$$(1+x)^5 = \binom{5}{0} + \binom{5}{1}x + \binom{5}{2}x^2 + \binom{5}{3}x^3 + \binom{5}{4}x^4 + \binom{5}{5}x^5$$

With $a = 1$, $b = x$, $n = 5$, this simplifies to $(1+x)^5 = \sum_{r=0}^5 \binom{5}{r} x^r$.

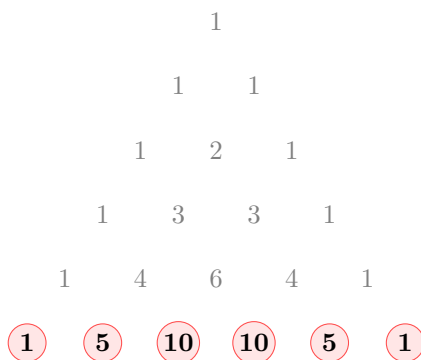
Step 2: Evaluate each term.

$$= 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$$

Evaluate the binomial coefficients: $\binom{5}{0} = 1$, $\binom{5}{1} = 5$, $\binom{5}{2} = 10$, $\binom{5}{3} = 10$, $\binom{5}{4} = 5$, $\binom{5}{5} = 1$.

Final answer: $1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$

Diagram.



Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 02 (be-002)

Difficulty: Foundation / Marks: 3

Expand $(x+3)^4$ in ascending powers of x .

Worked solution.

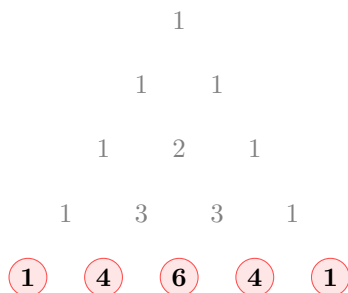
Apply the binomial theorem.

$$\begin{aligned}
 (x+3)^4 &= \binom{4}{0}(3)^4 + \binom{4}{1}(3)^3x + \binom{4}{2}(3)^2x^2 + \binom{4}{3}(3)x^3 + \binom{4}{4}x^4 \\
 &= 81 + 4(27)x + 6(9)x^2 + 4(3)x^3 + x^4 \\
 &= 81 + 108x + 54x^2 + 12x^3 + x^4
 \end{aligned}$$

The binomial theorem states $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Here $a = x$, $b = 3$, $n = 4$. Since the question asks for ascending powers of x , write the terms starting from the constant.

Final answer: $81 + 108x + 54x^2 + 12x^3 + x^4$

Diagram.



Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 03 (be-003)

Difficulty: Foundation / Marks: 3

Expand $(2 + x)^4$ fully, giving all terms in ascending powers of x .

Worked solution.

Step 1: Apply the binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$$(2 + x)^4 = \binom{4}{0}(2)^4 + \binom{4}{1}(2)^3x + \binom{4}{2}(2)^2x^2 + \binom{4}{3}(2)x^3 + \binom{4}{4}x^4$$

Write out the expansion with $a = 2$, $b = x$, $n = 4$.

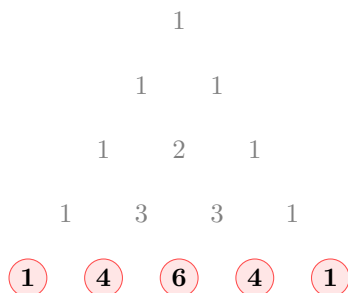
Step 2: Evaluate each term.

$$= 16 + 32x + 24x^2 + 8x^3 + x^4$$

Compute each term: 1×16 , 4×8 , 6×4 , 4×2 , 1×1 .

Final answer: $16 + 32x + 24x^2 + 8x^3 + x^4$

Diagram.



Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 04 (be-004)

Difficulty: Foundation / Marks: 4

Expand $(1 - 2x)^5$ in ascending powers of x , giving all terms.

Worked solution.

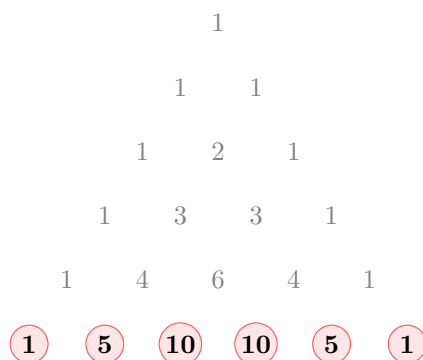
Apply the binomial theorem.

$$\begin{aligned}(1 - 2x)^5 &= \binom{5}{0} + \binom{5}{1}(-2x) + \binom{5}{2}(-2x)^2 + \binom{5}{3}(-2x)^3 + \binom{5}{4}(-2x)^4 + \binom{5}{5}(-2x)^5 \\&= 1 + 5(-2x) + 10(4x^2) + 10(-8x^3) + 5(16x^4) + (-32x^5) \\&= 1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5\end{aligned}$$

The binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Here $a = 1$, $b = -2x$, $n = 5$. Be careful with the signs — each power of $-2x$ alternates sign.

Final answer: $1 - 10x + 40x^2 - 80x^3 + 80x^4 - 32x^5$

Diagram.



Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 05 (be-005)

Difficulty: Foundation / Marks: 4

Expand $(3x - 1)^4$ in ascending powers of x .

Worked solution.

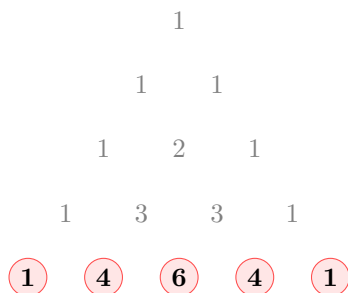
Apply the binomial theorem in ascending powers of x .

$$\begin{aligned}(3x - 1)^4 &= \binom{4}{0}(-1)^4 + \binom{4}{1}(-1)^3(3x) + \binom{4}{2}(-1)^2(3x)^2 + \binom{4}{3}(-1)(3x)^3 + \binom{4}{4}(3x)^4 \\&= 1 - 4(3x) + 6(9x^2) - 4(27x^3) + 81x^4 \\&= 1 - 12x + 54x^2 - 108x^3 + 81x^4\end{aligned}$$

The binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Here $a = -1$, $b = 3x$, $n = 4$. Writing $b = 3x$ as the ascending term ensures ascending powers of x .

Final answer: $1 - 12x + 54x^2 - 108x^3 + 81x^4$

Diagram.



Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 06 (be-006)

Difficulty: Foundation / Marks: 3

Expand $(2x + y)^3$ fully.

Worked solution.

Step 1: Apply the binomial theorem: $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$$(2x + y)^3 = \binom{3}{0}(2x)^3 + \binom{3}{1}(2x)^2(y) + \binom{3}{2}(2x)(y)^2 + \binom{3}{3}(y)^3$$

Apply the binomial theorem with $a = 2x$, $b = y$, $n = 3$.

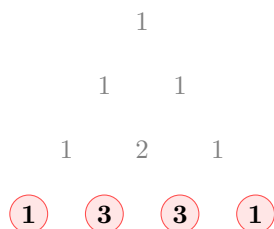
Step 2: Evaluate each term.

$$= 8x^3 + 12x^2y + 6xy^2 + y^3$$

Evaluate: $1 \times 8x^3 + 3 \times 4x^2y + 3 \times 2xy^2 + 1 \times y^3$.

Final answer: $8x^3 + 12x^2y + 6xy^2 + y^3$

Diagram.



Row 3 of Pascal's triangle (red) gives the binomial coefficients $\binom{3}{r}$; multiply each by the matching powers $a^{3-r}b^r$.

Binomial Expansion 07 (be-007)

Difficulty: Foundation / Marks: 4

Expand $(1 + 3x)^6$ in ascending powers of x , up to and including the term in x^3 .

Worked solution.

Step 1: Apply the binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$$(1 + 3x)^6 \approx \binom{6}{0} + \binom{6}{1}(3x) + \binom{6}{2}(3x)^2 + \binom{6}{3}(3x)^3$$

Write the first four terms of the expansion with $b = 3x$.

Step 2: Evaluate each term.

$$= 1 + 6(3x) + 15(9x^2) + 20(27x^3)$$

Compute: $\binom{6}{1} = 6$, $\binom{6}{2} = 15$, $\binom{6}{3} = 20$; and $(3x)^2 = 9x^2$, $(3x)^3 = 27x^3$.

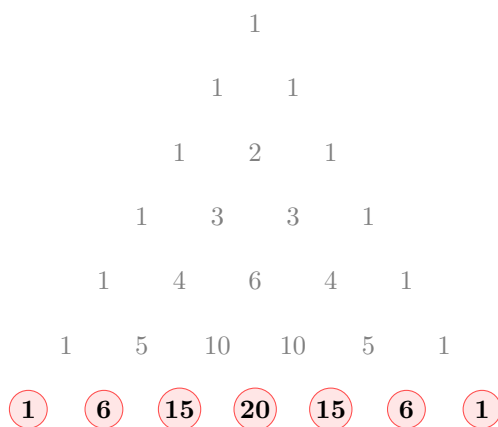
Step 3: Simplify.

$$= 1 + 18x + 135x^2 + 540x^3 + \dots$$

Simplify.

Final answer: $1 + 18x + 135x^2 + 540x^3 + \dots$

Diagram.



Row 6 of Pascal's triangle (red) gives the binomial coefficients $\binom{6}{r}$; multiply each by the matching powers $a^{6-r}b^r$.

Binomial Expansion 08 (be-008)

Difficulty: Foundation / Marks: 2

Without using a calculator, evaluate $\binom{7}{3}$.

Worked solution.

Step 1: Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{7}{3} = \frac{7!}{3! \times 4!} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1}$$

Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

Step 2: Simplify each term.

$$= \frac{210}{6} = 35$$

Cancel and simplify.

Final answer: $\binom{7}{3} = 35$

Diagram.

$$\binom{7}{3} = \frac{7!}{3!(7-3)!} = 35$$

The binomial coefficient counts the ways to choose r objects from 7; large factorials cancel.

Binomial Expansion 09 (be-009)

Difficulty: Foundation / Marks: 2

Without using a calculator, evaluate $\binom{9}{2}$.

Worked solution.

Step 1: Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{9}{2} = \frac{9!}{2! \times 7!} = \frac{9 \times 8}{2 \times 1}$$

Apply the formula.

Step 2: Simplify.

$$= \frac{72}{2} = 36$$

Simplify.

Final answer: $\binom{9}{2} = 36$

Diagram.

$$\binom{9}{2} = \frac{9!}{2!(9-2)!} = 36$$

The binomial coefficient counts the ways to choose r objects from 9; large factorials cancel.

Binomial Expansion 10 (be-010)

Difficulty: Foundation / Marks: 2

Without using a calculator, evaluate $\binom{10}{4}$.

Worked solution.

Step 1: Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{10}{4} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1}$$

Apply the formula.

Step 2: Evaluate each term.

$$= \frac{5040}{24} = 210$$

Compute numerator and denominator.

Final answer: $\binom{10}{4} = 210$

Diagram.

$$\binom{10}{4} = \frac{10!}{4!(10-4)!} = 210$$

The binomial coefficient counts the ways to choose r objects from 10; large factorials cancel.

Binomial Expansion 11 (be-011)

Difficulty: Foundation / Marks: 2

Without using a calculator, evaluate $\binom{8}{5}$.

Worked solution.

Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = \frac{336}{6} = 56$$

Use the symmetry property: $\binom{8}{5} = \binom{8}{3}$.

Final answer: $\binom{8}{5} = 56$

Diagram.

$$\binom{8}{5} = \frac{8!}{5!(8-5)!} = 56$$

The binomial coefficient counts the ways to choose r objects from 8; large factorials cancel.

Binomial Expansion 12 (be-012)

Difficulty: Foundation / Marks: 2

Evaluate $\binom{12}{3}$ without a calculator.

Worked solution.

Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{12}{3} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = \frac{1320}{6} = 220$$

Apply the formula.

Final answer: $\binom{12}{3} = 220$

Diagram.

$$\binom{12}{3} = \frac{12!}{3!(12-3)!} = 220$$

The binomial coefficient counts the ways to choose r objects from 12; large factorials cancel.

Binomial Expansion 13 (be-013)

Difficulty: Foundation / Marks: 2

Find the value of $\binom{n}{2}$ when $n = 6$.

Worked solution.

Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{6}{2} = \frac{6 \times 5}{2 \times 1} = \frac{30}{2} = 15$$

Substitute $n = 6$ into the formula.

Final answer: $\binom{6}{2} = 15$

Diagram.

$$\binom{6}{2} = \frac{6!}{2!(6-2)!} = 15$$

The binomial coefficient counts the ways to choose r objects from 6; large factorials cancel.

Binomial Expansion 14 (be-014)

Difficulty: Foundation / Marks: 3

Evaluate $\binom{11}{4}$ without a calculator, showing all working.

Worked solution.

Step 1: Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\binom{11}{4} = \frac{11!}{4! \times 7!} = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1}$$

Write out the formula.

Step 2: Evaluate each term.

$$\binom{11}{4} = \frac{7920}{24} = 330$$

Compute numerator: $11 \times 10 = 110$, $110 \times 9 = 990$, $990 \times 8 = 7920$. Denominator: $4! = 24$.

Final answer: $\binom{11}{4} = 330$

Diagram.

$$\binom{11}{4} = \frac{11!}{4!(11-4)!} = 330$$

The binomial coefficient counts the ways to choose r objects from 11; large factorials cancel.

Binomial Expansion 15 (be-015)

Difficulty: Foundation / Marks: 2

Find the coefficient of x^3 in the expansion of $(1+x)^8$.

Worked solution.

Use the binomial theorem to identify the general term.

$$\text{Coefficient of } x^3 = \binom{8}{3} = \frac{8 \times 7 \times 6}{3!} = \frac{336}{6} = 56$$

By the binomial theorem, $(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$. The general term is $\binom{n}{r} x^r$. To find the coefficient of x^3 , set $r = 3$.

Final answer: 56

Diagram.

$$T_{r+1} = \binom{8}{r} (1)^{8-r} (x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1+x)^8$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 16 (be-016)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^4 in the expansion of $(1 - x)^{10}$.

Worked solution.

Step 1: The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\text{Term} = \binom{10}{4}(-1)^4x^4 = 210 \times 1 \times x^4$$

The general term is $\binom{10}{r}(-x)^r = \binom{10}{r}(-1)^r x^r$. For x^4 , set $r = 4$.

Step 2: Simplify each term.

$$\text{Coefficient of } x^4 = 210$$

The coefficient is therefore +210.

Final answer: 210

Diagram.

$$T_{r+1} = \binom{10}{r} (1)^{10-r} (-x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + -x)^{10}$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 17 (be-017)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^2 in the expansion of $(1 + 3x)^{11}$.

Worked solution.

The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\text{Term} = \binom{11}{2}(3x)^2 = 55 \times 9x^2 = 495x^2$$

General term: $\binom{11}{r}(3x)^r$. For x^2 , set $r = 2$.

Final answer: 495

Diagram.

$$T_{r+1} = \binom{11}{r} (1)^{11-r} (3x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + 3x)^{11}$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 18 (be-018)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^3 in the expansion of $(1 - 2x)^9$.

Worked solution.

The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\text{Term} = \binom{9}{3}(-2)^3x^3 = 84 \times (-8)x^3 = -672x^3$$

General term: $\binom{9}{r}(-2x)^r$. For x^3 , set $r = 3$.

Final answer: -672

Diagram.

$$T_{r+1} = \binom{9}{r} (1)^{9-r} (-2x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + -2x)^9$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 19 (be-019)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^2 in the expansion of $(2 + x)^6$.

Worked solution.

The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\text{Term} = \binom{6}{2}(2)^4x^2 = 15 \times 16 \times x^2 = 240x^2$$

General term: $\binom{6}{r}(2)^{6-r}x^r$. For x^2 , set $r = 2$.

Final answer: 240

Diagram.

$$T_{r+1} = \binom{6}{r} (2)^{6-r} (x)^r$$

choose r power of 1st term power of 2nd term

General term of $(2 + x)^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 20 (be-020)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^3 in the expansion of $(3 + 2x)^5$.

Worked solution.

The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\text{Term} = \binom{5}{3}(3)^2(2)^3x^3 = 10 \times 9 \times 8 \times x^3 = 720x^3$$

General term: $\binom{5}{r}(3)^{5-r}(2x)^r$. For x^3 , set $r = 3$.

Final answer: 720

Diagram.

$$T_{r+1} = \binom{5}{r} (3)^{5-r} (2x)^r$$

choose r power of 1st term power of 2nd term

General term of $(3 + 2x)^5$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 21 (be-021)

Difficulty: Foundation / Marks: 2

Find the coefficient of x^5 in the expansion of $(1 + x)^7$.

Worked solution.

The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\binom{7}{2} = \frac{7 \times 6}{2} = 21$$

General term: $\binom{7}{r}x^r$. For x^5 , set $r = 5$. Use symmetry: $\binom{7}{5} = \binom{7}{2}$.

Final answer: 21

Diagram.

$$T_{r+1} = \binom{7}{r} (1)^{7-r} (x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + x)^7$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 22 (be-022)

Difficulty: Foundation / Marks: 4

The expansion of $(1 + ax)^6$ has coefficient of x^2 equal to 60. Find the value of a .

Worked solution.

Step 1: The general term of $(a + b)^n$ is $\binom{n}{r}a^{n-r}b^r$. Identify the term with the required power.

$$\binom{6}{2}a^2 = 15a^2$$

The coefficient of x^2 in $(1 + ax)^6$ is $\binom{6}{2}a^2$.

Step 2: Set the power of x to zero.

$$15a^2 = 60 \implies a^2 = 4 \implies a = \pm 2$$

Set this equal to 60 and solve.

Final answer: $a = 2$ or $a = -2$

Diagram.

$$\binom{6}{2}a^2 = 60 \xrightarrow{\text{coeff of } x^2} 15a^2 = 60 \Rightarrow a^2 = 4 \xrightarrow{\text{solve}} a = \pm 2$$

Solution flow: each arrow applies one step.

Binomial Expansion 23 (be-023)

Difficulty: Foundation / Marks: 4

Find the term independent of x in the expansion of $\left(x - \frac{1}{x}\right)^6$.

Worked solution.

Step 1: Write the general term using the binomial theorem.

$$T_{r+1} = \binom{6}{r}(x)^{6-r}\left(-\frac{1}{x}\right)^r = \binom{6}{r}(-1)^r x^{6-2r}$$

By the binomial theorem, $(a + b)^n = \sum_{r=0}^n \binom{n}{r}a^{n-r}b^r$. Here $a = x$, $b = -\frac{1}{x}$, $n = 6$. Combine the powers of x into a single exponent.

Step 2: Set the power of x to zero to find the term independent of x .

$$6 - 2r = 0 \implies r = 3$$

The term independent of x is the term where the exponent of x equals zero.

Step 3: Substitute $r = 3$ into the general term.

$$T_4 = \binom{6}{3}(-1)^3 = 20 \times (-1) = -20$$

Final answer: -20

Diagram.

$$T_{r+1} = \binom{6}{r} (x)^{6-r} \left(-\frac{1}{x}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $(x + -\frac{1}{x})^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 24 (be-024)

Difficulty: Foundation / Marks: 4

Find the term independent of x in the expansion of $\left(x + \frac{2}{x}\right)^4$.

Worked solution.

Step 1: Write the general term using $(a + b)^n = \sum \binom{n}{r} a^{n-r} b^r$ and set the power of x to zero.

$$T_{r+1} = \binom{4}{r} 2^r x^{4-2r}$$

General term: $\binom{4}{r} (x)^{4-r} \left(\frac{2}{x}\right)^r = \binom{4}{r} 2^r x^{4-r} x^{-r} = \binom{4}{r} 2^r x^{4-2r}$.

Step 2: Set the power of x to zero.

$$r = 2$$

Set $4 - 2r = 0 \Rightarrow r = 2$.

Step 3: Substitute and evaluate.

$$T_3 = 24$$

Substitute: $\binom{4}{2} \times 2^2 = 6 \times 4 = 24$.

Final answer: 24

Diagram.

$$T_{r+1} = \binom{4}{r} (x)^{4-r} \left(\frac{2}{x}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $(x + \frac{2}{x})^4$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 25 (be-025)

Difficulty: Foundation / Marks: 5

Find the term independent of x in the expansion of $\left(2x - \frac{1}{x^2}\right)^6$.

Worked solution.

Step 1: Write the general term using $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$ and set the power of x to zero.

$$T_{r+1} = \binom{6}{r} 2^{6-r} (-1)^r x^{6-3r}$$

$$\text{General term: } \binom{6}{r} (2x)^{6-r} \left(-\frac{1}{x^2}\right)^r = \binom{6}{r} 2^{6-r} (-1)^r x^{6-r} \cdot x^{-2r}.$$

Step 2: Find the required term.

$$r = 2$$

For independence from x : $6 - 3r = 0 \Rightarrow r = 2$.

Step 3: Substitute and evaluate.

$$T_3 = 240$$

$$\text{Substitute: } \binom{6}{2} \times 2^4 \times (-1)^2 = 15 \times 16 \times 1 = 240.$$

Final answer: 240

Diagram.

$$T_{r+1} = \binom{6}{r} (2x)^{6-r} \left(-\frac{1}{x^2}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $\left(2x + -\frac{1}{x^2}\right)^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 26 (be-026)

Difficulty: Foundation / Marks: 4

Find the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Worked solution.

Step 1: Write the general term using $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$ and set the power of x to zero.

$$T_{r+1} = \binom{6}{r} x^{12-3r}$$

$$\text{General term: } \binom{6}{r} (x^2)^{6-r} \left(\frac{1}{x}\right)^r = \binom{6}{r} x^{12-2r} \cdot x^{-r} = \binom{6}{r} x^{12-3r}.$$

Step 2: Set the power of x to zero.

$$r = 4$$

Set $12 - 3r = 0 \Rightarrow r = 4$.

Step 3: Substitute and evaluate.

$$T_5 = 15$$

$$\text{Substitute: } \binom{6}{4} = \binom{6}{2} = 15.$$

Final answer: 15

Diagram.

$$T_{r+1} = \binom{6}{r} (x^2)^{6-r} \left(\frac{1}{x}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $(x^2 + \frac{1}{x})^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 27 (be-027)

Difficulty: Foundation / Marks: 5

Find the term independent of x in the expansion of $\left(x - \frac{3}{x}\right)^8$.

Worked solution.

Step 1: Write the general term using $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$ and set the power of x to zero.

$$T_{r+1} = \binom{8}{r} (-3)^r x^{8-2r}$$

General term: $\binom{8}{r} (x)^{8-r} \left(-\frac{3}{x}\right)^r = \binom{8}{r} (-3)^r x^{8-2r}$.

Step 2: Set the power of x to zero.

$$r = 4$$

Set $8 - 2r = 0 \Rightarrow r = 4$.

Step 3: Substitute and evaluate.

$$T_5 = 5670$$

Substitute: $\binom{8}{4} \times (-3)^4 = 70 \times 81 = 5670$.

Final answer: 5670

Diagram.

$$T_{r+1} = \binom{8}{r} (x)^{8-r} \left(-\frac{3}{x}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $(x + \frac{3}{x})^8$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 28 (be-028)

Difficulty: Foundation / Marks: 5

Find the term independent of x in the expansion of $\left(3x + \frac{1}{x^2}\right)^9$.

Worked solution.

Step 1: Write the general term using $(a + b)^n = \sum \binom{n}{r} a^{n-r} b^r$ and set the power of x to zero.

$$T_{r+1} = \binom{9}{r} 3^{9-r} x^{9-3r}$$

General term: $\binom{9}{r} (3x)^{9-r} \left(\frac{1}{x^2}\right)^r = \binom{9}{r} 3^{9-r} x^{9-r} x^{-2r} = \binom{9}{r} 3^{9-r} x^{9-3r}$.

Step 2: Set the power of x to zero.

$$r = 3$$

Set $9 - 3r = 0 \Rightarrow r = 3$.

Step 3: Substitute and evaluate.

$$T_4 = 61236$$

Substitute: $\binom{9}{3} \times 3^6 = 84 \times 729 = 61236$.

Final answer: 61236

Diagram.

$$T_{r+1} = \binom{9}{r} (3x)^{9-r} \left(\frac{1}{x^2}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $\left(3x + \frac{1}{x^2}\right)^9$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 29 (be-029)

Difficulty: Foundation / Marks: 7

- (a) Write down the first three terms, in ascending powers of x , of the expansion of $(1 + x)^{10}$.
- (b) Use your expansion with $x = 0.01$ to find an approximate value of 1.01^{10} .
- (c) Calculate the percentage error in your approximation compared to the exact value $1.01^{10} = 1.10462$ (to 5 d.p.).

Worked solution.

Step 1: Apply the binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Write the first few terms.

$$(1 + x)^{10} \approx 1 + 10x + \binom{10}{2} x^2 = 1 + 10x + 45x^2$$

- (a) The first three terms are:

Step 2: Simplify each term.

$$1 + 10(0.01) + 45(0.01)^2 = 1 + 0.1 + 45 \times 0.0001 = 1 + 0.1 + 0.0045 = 1.1045$$

- (b) Substitute $x = 0.01$:

Step 3: Collect and simplify.

$$\% \text{ error} = \frac{|1.1045 - 1.10462|}{1.10462} \times 100 = \frac{0.00012}{1.10462} \times 100 \approx 0.011\%$$

- (c) Percentage error:

Final answer: (a) $1 + 10x + 45x^2$; (b) 1.1045; (c) approximately 0.011%

Diagram.

$$1 + 10x + 45x^2 \xrightarrow{\text{first 3 terms}} x = 0.01 : 1 + 0.1 + 0.0045 \xrightarrow{\text{substitute}} \approx 1.1045$$

Solution flow: each arrow applies one step.

Binomial Expansion 30 (be-030)

Difficulty: Foundation / Marks: 6

(a) Expand $(2 - x)^5$ in ascending powers of x , up to and including the x^3 term.

(b) Use $x = 0.1$ to find an approximate value of 1.9^5 .

Worked solution.

Step 1: Apply the binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$$(2 - x)^5 = \binom{5}{0} 2^5 + \binom{5}{1} 2^4 (-x) + \binom{5}{2} 2^3 (-x)^2 + \binom{5}{3} 2^2 (-x)^3 + \dots$$

(a) Write out the first four terms with $a = 2$, $b = -x$:

Step 2: Evaluate each term.

$$= 32 - 80x + 80x^2 - 40x^3 + \dots$$

Compute each term:

Step 3: Collect and simplify.

$$1.9^5 \approx 32 - 80(0.1) + 80(0.01) - 40(0.001) = 32 - 8 + 0.8 - 0.04 = 24.76$$

(b) Substitute $x = 0.1$:

Final answer: (a) $32 - 80x + 80x^2 - 40x^3$; (b) $1.9^5 \approx 24.76$

Diagram.

$$\begin{array}{ccccccc}
 & & & & 1 & & & & \\
 & & & & & & 1 & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & & & 1 & & 3 & & 3 & & 1 \\
 & & & & 1 & & 4 & & 6 & & 4 & & 1 \\
 & & & & & 1 & & 5 & & 10 & & 10 & & 5 & & 1
 \end{array}$$

Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 31 (be-031)

Difficulty: Foundation / Marks: 7

- (a) Write down the first four terms, in ascending powers of x , of the expansion of $(1-3x)^7$.
 (b) Use your expansion with a suitable value of x to estimate 0.97^7 , giving your answer to 4 decimal places.

Worked solution.

Step 1: Apply the binomial theorem: $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Write the first few terms.

$$(1-3x)^7 \approx 1 - 21x + 189x^2 - 945x^3 + \dots$$

- (a) The general term is $\binom{7}{r}(-3x)^r$. Write the first four terms:

Step 2: Simplify each term.

$$0.97^7 \approx 1 - 21(0.01) + 189(0.0001) - 945(0.000001)$$

- (b) We need 0.97^7 . Set $1-3x = 0.97 \Rightarrow x = 0.01$.

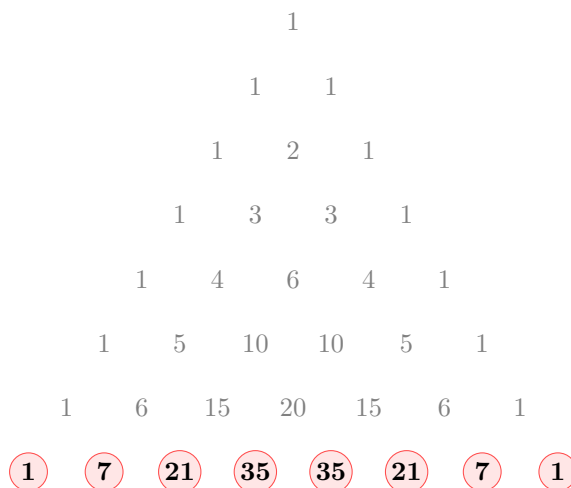
Step 3: Collect and simplify.

$$= 1 - 0.21 + 0.0189 - 0.000945 = 0.8080 \text{ (to 4 d.p.)}$$

Compute:

Final answer: (a) $1 - 21x + 189x^2 - 945x^3 + \dots$; (b) $0.97^7 \approx 0.8080$

Diagram.



Row 7 of Pascal's triangle (red) gives the binomial coefficients $\binom{7}{r}$; multiply each by the matching powers $a^{7-r}b^r$.

Binomial Expansion 32 (be-032)

Difficulty: Foundation / Marks: 8

- (a) Expand $(1+2x)^5$ fully.

(b) By substituting $x = 0.01$, use your expansion to estimate 1.02^5 .

(c) The exact value of 1.02^5 is 1.10408 (to 5 s.f.). Find the percentage error in your estimate.

Worked solution.

Step 1: Apply the binomial theorem: $(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

$$(1 + 2x)^5 = 1 + 10x + 40x^2 + 80x^3 + 80x^4 + 32x^5$$

(a) Use $a = 1$, $b = 2x$, $n = 5$:

Step 2: Simplify each term.

$$\approx 1 + 10(0.01) + 40(0.0001) + 80(0.000001) = 1 + 0.1 + 0.004 + 0.00008 = 1.10408$$

(b) Substitute $x = 0.01$ (using all terms up to x^3 for the estimate):

Step 3: Collect and simplify.

$$\% \text{ error} = \frac{|1.10408 - 1.10408|}{1.10408} \times 100 \approx 0\%$$

(c) Percentage error:

Step 4: Final simplification.

$$\text{Percentage error} < 0.001\%$$

In this case the approximation is very close (essentially exact to 5 s.f.) because the higher power terms are negligible.

Final answer: (a) $1 + 10x + 40x^2 + 80x^3 + 80x^4 + 32x^5$; (b) ≈ 1.1041 ; (c) *verysmall*, $< 0.001\%$

Diagram.

$$\begin{array}{ccccccc}
 & & & & 1 & & & & \\
 & & & & & 1 & & 1 & \\
 & & & & & & 1 & & 2 & & 1 \\
 & & & & & & & 1 & & 3 & & 3 & & 1 \\
 & & & & & & & & 1 & & 4 & & 6 & & 4 & & 1 \\
 & & & & & & & & & \textcircled{1} & & \textcircled{5} & & \textcircled{10} & & \textcircled{10} & & \textcircled{5} & & \textcircled{1}
 \end{array}$$

Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 33 (be-033)

Difficulty: Foundation / Marks: 8

(a) Write down the first four terms, in ascending powers of x , of the binomial expansion of $(1 + x)^{12}$.

(b) Hence, or otherwise, find the first four terms of the expansion of $(1 + x + x^2)(1 + x)^{12}$.

Worked solution.

Step 1: Use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$(1+x)^{12} \approx 1 + 12x + 66x^2 + 220x^3 + \dots$$

(a) The first four terms are:

Step 2: Simplify each term.

$$1 \times (1 + 12x + 66x^2 + 220x^3) + x \times (1 + 12x + 66x^2) + x^2 \times (1 + 12x)$$

(b) Multiply $(1+x+x^2)$ by $(1+12x+66x^2+220x^3+\dots)$. Collect terms up to x^3 :

Step 3: Collect and simplify.

$$= (1) + (12+1)x + (66+12+1)x^2 + (220+66+12)x^3 + \dots$$

Combine like terms:

Step 4: Final simplification.

$$= 1 + 13x + 79x^2 + 298x^3 + \dots$$

Simplify:

$$\text{Final answer: } (a)1 + 12x + 66x^2 + 220x^3; (b)1 + 13x + 79x^2 + 298x^3 + \dots$$

Diagram.

$$(1+x)^{12} = 1 + 12x + 66x^2 + 220x^3 \xrightarrow{\text{expand}} x(1+x+x^2) \xrightarrow{\text{multiply \& collect}} 1 + 13x + 79x^2 + 298x^3$$

Solution flow: each arrow applies one step.

Binomial Expansion 34 (be-034)

Difficulty: Foundation / Marks: 8

- (a) Find the first four terms, in ascending powers of x , of the expansion of $(2+x)^6$.
- (b) Use your expansion to show that $(2+x)^6 \approx 64 + 192x + 240x^2 + 160x^3$ for small x .
- (c) By substituting $x = 0.1$, use the expansion to estimate 2.1^6 to 3 significant figures.

Worked solution.

Step 1: Apply the binomial theorem: $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Write the first few terms.

$$(2+x)^6 = 64 + 6 \times 32x + 15 \times 16x^2 + 20 \times 8x^3 + \dots$$

(a) General term: $\binom{6}{r} 2^{6-r} x^r$. First four terms ($r = 0, 1, 2, 3$):

Step 2: Simplify.

$$= 64 + 192x + 240x^2 + 160x^3 + \dots$$

Simplify:

Step 3: Collect and simplify. (b) This confirms the four-term approximation for small x . ✓

Step 4: Final simplification.

$$2.1^6 \approx 64 + 192(0.1) + 240(0.01) + 160(0.001) = 64 + 19.2 + 2.4 + 0.16 = 85.76$$

(c) Substitute $x = 0.1$:

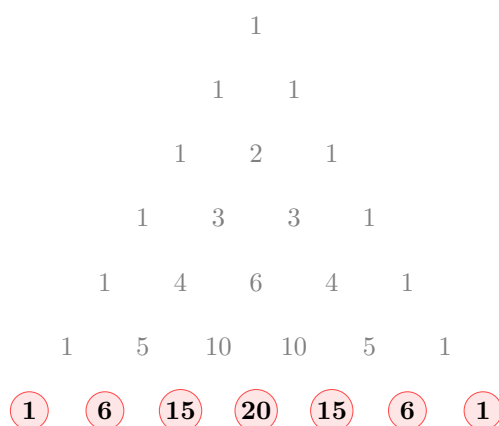
Step 5: Final answer.

$$2.1^6 \approx 85.8$$

Round to 3 significant figures:

Final answer: (a) $64 + 192x + 240x^2 + 160x^3$; (c) $2.1^6 \approx 85.8$

Diagram.



Row 6 of Pascal's triangle (red) gives the binomial coefficients $\binom{6}{r}$; multiply each by the matching powers $a^{6-r}b^r$.

Binomial Expansion 35 (be-035)

Difficulty: Foundation / Marks: 9

- (a) Write down the first four terms, in ascending powers of x , of the expansion of $(1-x)^8$.
 (b) Hence find the first four terms of the expansion of $(3+x)(1-x)^8$.
 (c) Find the coefficient of x^2 in the expansion of $(3+x)(1-x)^8$.

Worked solution.

Step 1: Apply the binomial theorem: $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$. Write the first few terms.

$$(1-x)^8 = 1 - 8x + 28x^2 - 56x^3 + \dots$$

- (a) Use $b = -x$, $n = 8$:

Step 2: Simplify each term.

$$3(1 - 8x + 28x^2 - 56x^3) + x(1 - 8x + 28x^2)$$

- (b) Multiply $(3+x)$ by $(1 - 8x + 28x^2 - 56x^3 + \dots)$:

Step 3: Collect and simplify.

$$= 3 + (-24 + 1)x + (84 - 8)x^2 + (-168 + 28)x^3 + \dots$$

Collect terms:

Step 4: Final simplification.

$$= 3 - 23x + 76x^2 - 140x^3 + \dots$$

Simplify:

Step 5: Final answer.

Coefficient of $x^2 = 76$

(c) The coefficient of x^2 is 76.

Final answer: (a) $1 - 8x + 28x^2 - 56x^3$; (b) $3 - 23x + 76x^2 - 140x^3 + \dots$; (c) 76

Diagram.

$$(1 - x)^8 = 1 - 8x + 28x^2 - 56x^3 \xrightarrow{\text{expand}} \times (3 + x) \xrightarrow{\text{multiply}} 3 - 23x + 76x^2 - 140x^3$$

Solution flow: each arrow applies one step.

Binomial Expansion 36 (be-036)

Difficulty: Foundation / Marks: 3

Expand $(1 + 2x)^4$ in ascending powers of x , up to and including the term in x^3 .

Worked solution.

Step 1: Apply the binomial theorem

$$\binom{4}{0} + \binom{4}{1}(2x) + \binom{4}{2}(2x)^2 + \binom{4}{3}(2x)^3$$

Use $(1 + 2x)^4$ with $b = 2x$, $n = 4$.

Step 2: Simplify

$$1 + 8x + 24x^2 + 32x^3$$

Evaluate each term.

Final answer: $1 + 8x + 24x^2 + 32x^3$

Diagram.

$$\begin{array}{ccccccc}
 & & & & 1 & & & & \\
 & & & & & 1 & & 1 & \\
 & & & & & & 1 & & 1 \\
 & & & & 1 & & 2 & & 1 \\
 & & & & & 1 & & 3 & & 1 \\
 & & & & & & 1 & & 3 & & 1 \\
 & & & & & & & 1 & & 3 & & 1 \\
 & & & & & & & & 1 & & 4 & & 1
 \end{array}$$

Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 37 (be-037)

Difficulty: Foundation / Marks: 3

Expand $(1 - 3x)^5$ in ascending powers of x , up to and including the term in x^3 .

Worked solution.

Step 1: Apply the binomial theorem

$$\binom{5}{0} + \binom{5}{1}(-3x) + \binom{5}{2}(-3x)^2 + \binom{5}{3}(-3x)^3$$

Use $(1 + (-3x))^5$.

Step 2: Simplify

$$1 - 15x + 90x^2 - 270x^3$$

Evaluate coefficients and signs.

Final answer: $1 - 15x + 90x^2 - 270x^3$

Diagram.

$$\begin{array}{ccccccc} & & & & 1 & & & & \\ & & & & & 1 & & 1 & \\ & & & & & & 1 & & 1 \\ & & & & 1 & & 2 & & 1 \\ & & & 1 & & 3 & & 3 & & 1 \\ & & 1 & & 4 & & 6 & & 4 & & 1 \\ & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 38 (be-038)

Difficulty: Foundation / Marks: 4

Find the first 4 terms in the expansion of $(2 + x)^6$ in ascending powers of x .

Worked solution.

Step 1: Apply the binomial theorem

$$\binom{6}{0}2^6 + \binom{6}{1}2^5x + \binom{6}{2}2^4x^2 + \binom{6}{3}2^3x^3$$

With $a = 2, b = x, n = 6$.

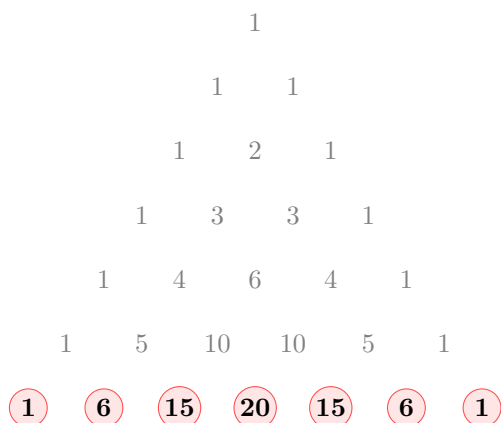
Step 2: Simplify

$$64 + 192x + 240x^2 + 160x^3$$

Evaluate each term.

Final answer: $64 + 192x + 240x^2 + 160x^3$

Diagram.



Row 6 of Pascal's triangle (red) gives the binomial coefficients $\binom{6}{r}$; multiply each by the matching powers $a^{6-r}b^r$.

Binomial Expansion 39 (be-039)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^3 in the expansion of $(1 + 4x)^5$.

Worked solution.

Step 1: General term

$$\binom{5}{3} \times 4^3 \times x^3 = 10 \times 64 \times x^3$$

The term in x^3 is $\binom{5}{3}(4x)^3$.

Step 2: Evaluate

$$640x^3$$

Multiply.

Final answer: 640

Diagram.

$$T_{r+1} = \binom{5}{r} (1)^{5-r} (4x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + 4x)^5$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 40 (be-040)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^2 in the expansion of $(3 - x)^6$.

Worked solution.

Step 1: General term for x^2

$$\binom{6}{2} \times 81 \times x^2 = 15 \times 81 \times x^2$$

The x^2 term is $\binom{6}{2}(3)^4(-x)^2$.

Step 2: Evaluate

$$1215x^2$$

Multiply.

Final answer: 1215

Diagram.

$$T_{r+1} = \binom{6}{r} (3)^{6-r} (-x)^r$$

choose r power of 1st term power of 2nd term

General term of $(3 - x)^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 41 (be-041)

Difficulty: Foundation / Marks: 3

Expand $(1 + x)^8$ up to and including the term in x^3 .

Worked solution.

Step 1: First 4 terms

$$\binom{8}{0} + \binom{8}{1}x + \binom{8}{2}x^2 + \binom{8}{3}x^3$$

Apply the binomial theorem with $n = 8$.

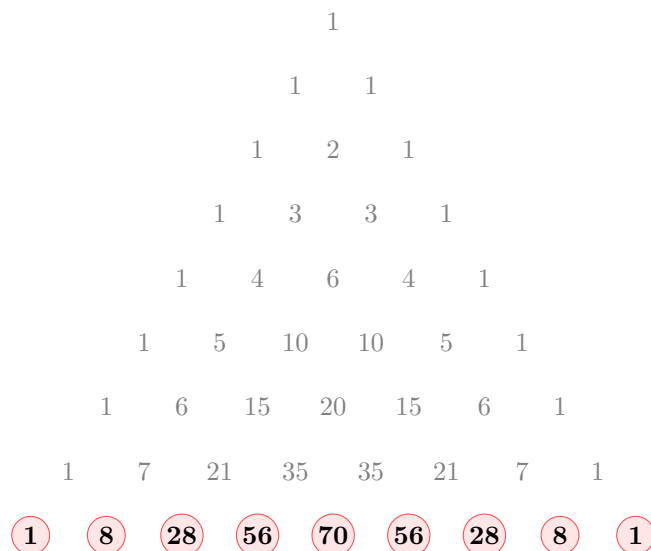
Step 2: Evaluate

$$1 + 8x + 28x^2 + 56x^3$$

Compute coefficients.

Final answer: $1 + 8x + 28x^2 + 56x^3$

Diagram.



Row 8 of Pascal's triangle (red) gives the binomial coefficients $\binom{8}{r}$; multiply each by the matching powers $a^{8-r}b^r$.

Binomial Expansion 42 (be-042)

Difficulty: Foundation / Marks: 3

Expand $(1 - 2x)^6$ up to and including the term in x^3 .

Worked solution.

Step 1: Substitute $b = -2x$

$$\binom{6}{0} + \binom{6}{1}(-2x) + \binom{6}{2}(-2x)^2 + \binom{6}{3}(-2x)^3$$

Apply binomial theorem.

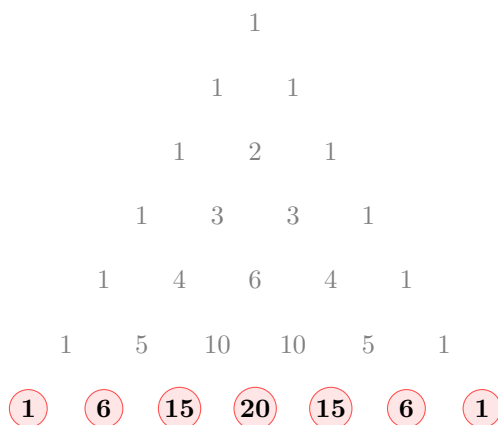
Step 2: Simplify

$$1 - 12x + 60x^2 - 160x^3$$

Evaluate each term.

Final answer: $1 - 12x + 60x^2 - 160x^3$

Diagram.



Row 6 of Pascal's triangle (red) gives the binomial coefficients $\binom{6}{r}$; multiply each by the matching powers $a^{6-r}b^r$.

Binomial Expansion 43 (be-043)

Difficulty: Foundation / Marks: 4

Find the first 4 terms of $(3 + 2x)^5$ in ascending powers of x .

Worked solution.

Step 1: Apply binomial theorem

$$\binom{5}{0}3^5 + \binom{5}{1}3^4(2x) + \binom{5}{2}3^3(2x)^2 + \binom{5}{3}3^2(2x)^3$$

With $a = 3, b = 2x, n = 5$.

Step 2: Evaluate

$$243 + 810x + 1080x^2 + 720x^3$$

Compute each term.

Final answer: $243 + 810x + 1080x^2 + 720x^3$

Diagram.

$$\begin{array}{ccccccc}
 & & & & 1 & & & & \\
 & & & & 1 & & 1 & & \\
 & & & 1 & & 2 & & 1 & \\
 & & 1 & & 3 & & 3 & & 1 \\
 & 1 & & 4 & & 6 & & 4 & & 1 \\
 \textcircled{1} & \textcircled{5} & \textcircled{10} & \textcircled{10} & \textcircled{5} & \textcircled{1}
 \end{array}$$

Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 44 (be-044)

Difficulty: Foundation / Marks: 2

Find the coefficient of x^4 in the expansion of $(1 + x)^{10}$.

Worked solution.

Use the general term

$$\binom{10}{4} = \frac{10!}{4! \times 6!} = 210$$

Coefficient of x^4 is $\binom{10}{4}$.

Final answer: 210

Diagram.

$$T_{r+1} = \binom{10}{r} (1)^{10-r} (x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1+x)^{10}$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 45 (be-045)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^2 in the expansion of $(1-5x)^4$.

Worked solution.

x^2 term

$$\binom{4}{2} \times 25x^2 = 6 \times 25 = 150x^2$$

The term is $\binom{4}{2}(-5x)^2$.

Final answer: 150

Diagram.

$$T_{r+1} = \binom{4}{r} (1)^{4-r} (-5x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1-5x)^4$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 46 (be-046)

Difficulty: Foundation / Marks: 4

Expand $(x-2)^5$ fully.

Worked solution.

Step 1: Apply binomial theorem

$$\sum_{r=0}^5 \binom{5}{r} x^{5-r} (-2)^r$$

With $a = x, b = -2, n = 5$.

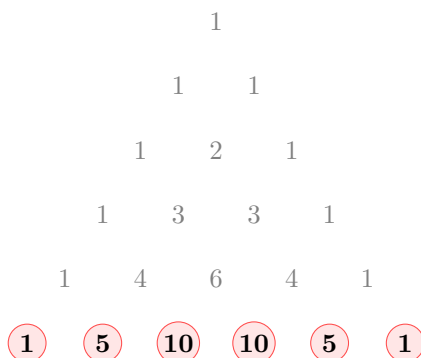
Step 2: Expand all terms

$$x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$$

Evaluate each term.

Final answer: $x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

Diagram.



Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 47 (be-047)

Difficulty: Foundation / Marks: 4

Find the term independent of x in the expansion of $(x + \frac{2}{x})^6$.

Worked solution.

Step 1: General term

$$x^{6-2r} = x^0 \implies r = 3$$

The general term is $\binom{6}{r}x^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^rx^{6-2r}$.

Step 2: Evaluate

$$\binom{6}{3} \times 2^3 = 20 \times 8 = 160$$

Substitute $r = 3$.

Final answer: 160

Diagram.

$$T_{r+1} = \binom{6}{r} (x)^{6-r} \left(\frac{2}{x}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $(x + \frac{2}{x})^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 48 (be-048)

Difficulty: Foundation / Marks: 3

In the expansion of $(1 + kx)^6$, the coefficient of x^2 is 60. Find the value of k , where $k > 0$.

Worked solution.*Step 1: x^2 coefficient*

$$15k^2 = 60$$

The coefficient is $\binom{6}{2}k^2 = 15k^2$.

Step 2: Solve

$$k^2 = 4 \implies k = 2$$

Divide by 15 and take the positive root.

Final answer: $k = 2$

Diagram.

$$\binom{6}{2}k^2 = 60 \xrightarrow{\text{coeff of } x^2} 15k^2 = 60 \xrightarrow{\text{solve } (k > 0)} k = 2$$

Solution flow: each arrow applies one step.

Binomial Expansion 49 (be-049)

Difficulty: Foundation / Marks: 3

Find the first 3 terms of $(1 + 3x)^{10}$ in ascending powers of x .

Worked solution.*Step 1: Apply binomial theorem*

$$1 + \binom{10}{1}(3x) + \binom{10}{2}(3x)^2$$

First 3 terms with $n = 10, b = 3x$.

Step 2: Evaluate

$$1 + 30x + 405x^2$$

Compute coefficients.

Final answer: $1 + 30x + 405x^2$

Diagram.

$$T_{r+1} = \binom{10}{r} (1)^{10-r} (3x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + 3x)^{10}$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 50 (be-050)

Difficulty: Foundation / Marks: 4

Expand $(2x - 1)^4$ fully.

Worked solution.

Step 1: Apply binomial theorem

$$(2x)^4 + \binom{4}{1}(2x)^3(-1) + \binom{4}{2}(2x)^2(-1)^2 + \binom{4}{3}(2x)(-1)^3 + (-1)^4$$

With $a = 2x, b = -1, n = 4$.

Step 2: Simplify

$$16x^4 - 32x^3 + 24x^2 - 8x + 1$$

Evaluate.

Final answer: $16x^4 - 32x^3 + 24x^2 - 8x + 1$

Diagram.

$$\begin{array}{ccccccc} & & & & 1 & & & & \\ & & & & & 1 & & 1 & \\ & & & & & & 1 & & 1 \\ & & & & 1 & & 2 & & 1 \\ & & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 51 (be-051)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^3 in the expansion of $(2 - x)^7$.

Worked solution.

x^3 term

$$\binom{7}{3} \times 16 \times (-x^3) = 35 \times 16 \times (-1) = -560$$

The term is $\binom{7}{3}(2)^4(-x)^3$.

Final answer: -560

Diagram.

$$T_{r+1} = \binom{7}{r} (2)^{7-r} (-x)^r$$

choose r power of 1st term power of 2nd term

General term of $(2 + -x)^7$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 52 (be-052)

Difficulty: Foundation / Marks: 3

Expand $(1 + x)^7$ up to the term in x^4 .

Worked solution.

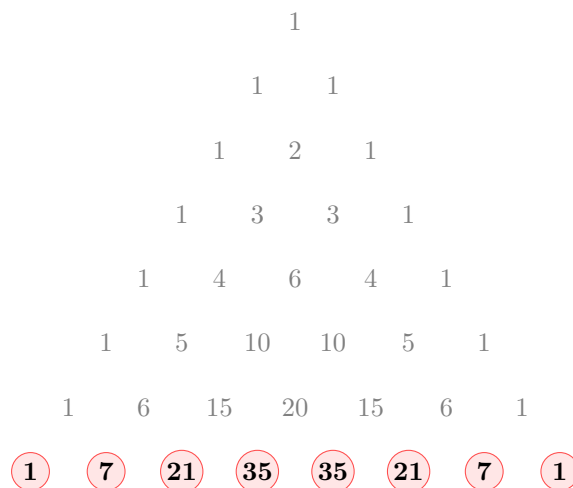
First 5 terms

$$1 + 7x + 21x^2 + 35x^3 + 35x^4$$

Apply binomial theorem.

Final answer: $1 + 7x + 21x^2 + 35x^3 + 35x^4$

Diagram.



Row 7 of Pascal's triangle (red) gives the binomial coefficients $\binom{7}{r}$; multiply each by the matching powers $a^{7-r}b^r$.

Binomial Expansion 53 (be-053)

Difficulty: Foundation / Marks: 3

Find the first 4 terms of $(1 - x)^{12}$ in ascending powers of x .

Worked solution.

Apply binomial theorem

$$1 - 12x + 66x^2 - 220x^3$$

With $b = -x, n = 12$.

Final answer: $1 - 12x + 66x^2 - 220x^3$

Diagram.

$$T_{r+1} = \binom{12}{r} (1)^{12-r} (-x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + -x)^{12}$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 54 (be-054)

Difficulty: Foundation / Marks: 3

In the expansion of $(1 + ax)^8$, the coefficient of x^3 is 1512. Find the value of a .

Worked solution.

Step 1: x^3 coefficient

$$56a^3 = 1512$$

The coefficient is $\binom{8}{3}a^3 = 56a^3$.

Step 2: Solve

$$a^3 = 27 \implies a = 3$$

Divide and take cube root.

Final answer: $a = 3$

Diagram.

$$\binom{8}{3}a^3 = 1512 \xrightarrow{\text{coeff of } x^3} 56a^3 = 1512 \Rightarrow a^3 = 27 \xrightarrow{\text{solve}} a = 3$$

Solution flow: each arrow applies one step.

Binomial Expansion 55 (be-055)

Difficulty: Foundation / Marks: 2

Find the coefficient of x^5 in $(1 + x)^9$.

Worked solution.

Evaluate

$$\binom{9}{5} = \frac{9!}{5! \times 4!} = 126$$

The coefficient is $\binom{9}{5}$.

Final answer: 126

Diagram.

$$T_{r+1} = \binom{9}{r} (1)^{9-r} (x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1+x)^9$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 56 (be-056)

Difficulty: Foundation / Marks: 3

Find the first 4 terms of $(5-x)^4$ in descending powers of 5.

Worked solution.

Step 1: Apply binomial theorem

$$5^4 + \binom{4}{1}5^3(-x) + \binom{4}{2}5^2(-x)^2 + \binom{4}{3}5(-x)^3$$

With $a = 5, b = -x, n = 4$.

Step 2: Simplify

$$625 - 500x + 150x^2 - 20x^3$$

Evaluate.

Final answer: $625 - 500x + 150x^2 - 20x^3$

Diagram.

$$\begin{array}{ccccccc} & & & & 1 & & & & \\ & & & & & 1 & & 1 & \\ & & & 1 & & 2 & & 1 & \\ & & 1 & & 3 & & 3 & & 1 \\ \textcircled{1} & & \textcircled{4} & & \textcircled{6} & & \textcircled{4} & & \textcircled{1} \end{array}$$

Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 57 (be-057)

Difficulty: Foundation / Marks: 4

Use the binomial expansion to find an approximation for $(1.02)^{10}$, using the first 3 terms of $(1+0.02)^{10}$.

Worked solution.

Step 1: First 3 terms

$$1 + 10(0.02) + 45(0.02)^2$$

Set $x = 0.02, n = 10$.

Step 2: Evaluate

$$1 + 0.2 + 0.018 = 1.218$$

Calculate each term.

Final answer: 1.218

Diagram.

$$1 + 10(0.02) + 45(0.02)^2 \xrightarrow{\text{first 3 terms}} 1 + 0.2 + 0.018 \xrightarrow{\text{evaluate}} \approx 1.218$$

Solution flow: each arrow applies one step.

Binomial Expansion 58 (be-058)

Difficulty: Foundation / Marks: 4

Use the binomial expansion to approximate $(0.98)^8$ using the first 3 terms of $(1 - 0.02)^8$.

Worked solution.

Step 1: First 3 terms

$$1 + 8(-0.02) + 28(-0.02)^2$$

Set $x = -0.02, n = 8$.

Step 2: Evaluate

$$1 - 0.16 + 0.0112 = 0.8512$$

Calculate.

Final answer: 0.8512

Diagram.

$$1 - 8(0.02) + 28(0.02)^2 \xrightarrow{\text{first 3 terms}} 1 - 0.16 + 0.0112 \xrightarrow{\text{evaluate}} \approx 0.8512$$

Solution flow: each arrow applies one step.

Binomial Expansion 59 (be-059)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^2 in $(3 + 2x)^5$.

Worked solution.

x^2 term

$$10 \times 27 \times 4x^2 = 1080x^2$$

The term is $\binom{5}{2}(3)^3(2x)^2$.

Final answer: 1080

Diagram.

$$T_{r+1} = \binom{5}{r} (3)^{5-r} (2x)^r$$

choose r power of 1st term power of 2nd term

General term of $(3 + 2x)^5$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 60 (be-060)

Difficulty: Foundation / Marks: 3

Expand $(1 + x)^6$ fully.

Worked solution.

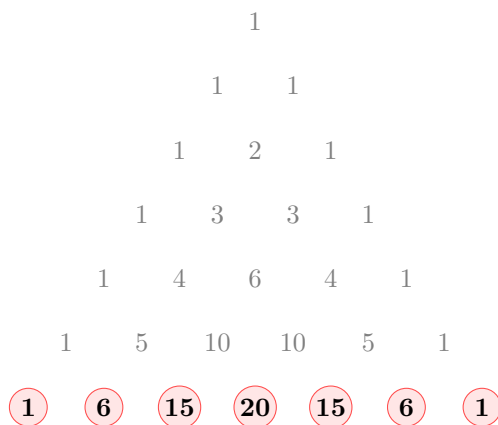
Full expansion

$$1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + x^6$$

Apply binomial theorem with $n = 6$.

Final answer: $1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + x^6$

Diagram.



Row 6 of Pascal's triangle (red) gives the binomial coefficients $\binom{6}{r}$; multiply each by the matching powers $a^{6-r}b^r$.

Binomial Expansion 61 (be-061)

Difficulty: Foundation / Marks: 3

In the expansion of $(2 + kx)^5$, the coefficient of x^2 is 80. Find k , where $k > 0$.

Worked solution.

Step 1: x^2 term

$$80k^2 = 80$$

The coefficient is $\binom{5}{2} \times 2^3 \times k^2 = 10 \times 8 \times k^2 = 80k^2$.

Step 2: Solve

$$k^2 = 1 \implies k = 1$$

Divide.

Final answer: $k = 1$

Diagram.

$$\binom{5}{2} 2^3 k^2 = 80 \xrightarrow{\text{coeff of } x^2} 80k^2 = 80 \xrightarrow{\text{solve}} k = 1$$

Solution flow: each arrow applies one step.

Binomial Expansion 62 (be-062)

Difficulty: Foundation / Marks: 3

Find the first 3 terms of $(1 + \frac{x}{2})^8$ in ascending powers of x .

Worked solution.

Step 1: Apply binomial theorem

$$1 + 8 \times \frac{x}{2} + 28 \times \frac{x^2}{4}$$

With $b = \frac{x}{2}, n = 8$.

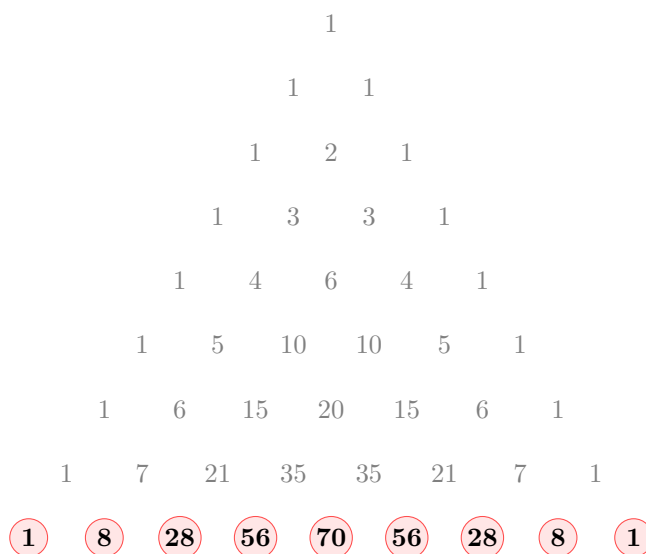
Step 2: Simplify

$$1 + 4x + 7x^2$$

Evaluate.

Final answer: $1 + 4x + 7x^2$

Diagram.



Row 8 of Pascal's triangle (red) gives the binomial coefficients $\binom{8}{r}$; multiply each by the matching powers $a^{8-r}b^r$.

Binomial Expansion 63 (be-063)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^4 in the expansion of $(1 - 2x)^8$.

Worked solution.

x^4 term

$$70 \times 16 = 1120$$

The term is $\binom{8}{4}(-2x)^4 = 70 \times 16x^4$.

Final answer: 1120

Diagram.

$$T_{r+1} = \binom{8}{r} (1)^{8-r} (-2x)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + -2x)^8$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 64 (be-064)

Difficulty: Foundation / Marks: 4

Use the expansion of $(1 + x)^5$ to estimate $(1.1)^5$ correct to 3 decimal places.

Worked solution.

Step 1: Expand

$$1 + 5(0.1) + 10(0.01) + 10(0.001) + 5(0.0001) + 0.00001$$

Set $x = 0.1$.

Step 2: Sum

$$1 + 0.5 + 0.1 + 0.01 + 0.0005 + 0.00001 = 1.61051$$

Add all terms.

Final answer: 1.611

Diagram.

$$1 + 5(0.1) + 10(0.1)^2 + 10(0.1)^3 + \dots \xrightarrow{\text{expand } (1 + 0.1)^5} 1 + 0.5 + 0.1 + 0.01 + \dots \xrightarrow{\text{evaluate}} \approx 1.611$$

Solution flow: each arrow applies one step.

Binomial Expansion 65 (be-065)

Difficulty: Foundation / Marks: 3

Expand $(3 - x)^4$ fully.

Worked solution.

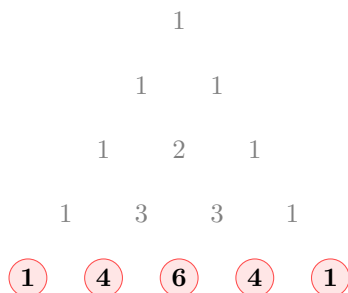
Apply binomial theorem

$$81 - 108x + 54x^2 - 12x^3 + x^4$$

With $a = 3, b = -x, n = 4$.

Final answer: $81 - 108x + 54x^2 - 12x^3 + x^4$

Diagram.



Row 4 of Pascal's triangle (red) gives the binomial coefficients $\binom{4}{r}$; multiply each by the matching powers $a^{4-r}b^r$.

Binomial Expansion 66 (be-066)

Difficulty: Foundation / Marks: 3

Find the coefficient of x^3 in $(1 + \frac{x}{3})^9$.

Worked solution.

x^3 term

$$\frac{84}{27} = \frac{28}{9}$$

The term is $\binom{9}{3} \left(\frac{x}{3}\right)^3 = 84 \times \frac{x^3}{27}$.

Final answer: $\frac{28}{9}$

Diagram.

$$T_{r+1} = \binom{9}{r} (1)^{9-r} \left(\frac{x}{3}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $(1 + \frac{x}{3})^9$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 67 (be-067)

Difficulty: Foundation / Marks: 4

Find the first 4 terms of $(4 - x)^5$ in ascending powers of x .

Worked solution.

Step 1: Apply binomial theorem

$$4^5 - \binom{5}{1} 4^4 x + \binom{5}{2} 4^3 x^2 - \binom{5}{3} 4^2 x^3 + \binom{5}{4} 4 x^4 - x^5$$

With $a = 4, b = -x, n = 5$.

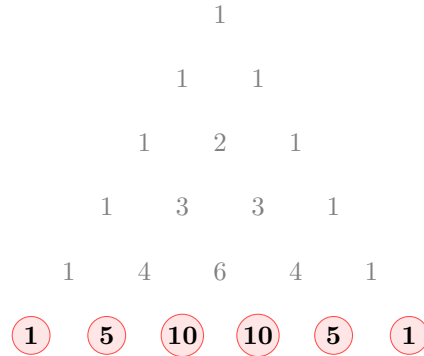
Step 2: Evaluate

$$1024 - 1280x + 640x^2 - 160x^3$$

Compute.

Final answer: $1024 - 1280x + 640x^2 - 160x^3$

Diagram.



Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

Binomial Expansion 68 (be-068)

Difficulty: Foundation / Marks: 4

The expansion of $(1 + px)^n$ begins $1 + 12x + 54x^2 + \dots$. Find p and n .

Worked solution.

Step 1: Set up equations

$$np = 12, \quad \frac{n(n-1)}{2}p^2 = 54$$

Coefficient of x : $np = 12$. Coefficient of x^2 : $\frac{n(n-1)}{2}p^2 = 54$.

Step 2: Solve

$$\frac{n(n-1)}{2} \times \frac{144}{n^2} = 54 \implies \frac{72(n-1)}{n} = 54$$

From first: $p = 12/n$. Substitute into second.

Step 3: Find n and p

$$72n - 72 = 54n \implies 18n = 72 \implies n = 4, ; p = 3$$

Solve for n .

Final answer: $n = 4, p = 3$

Diagram.

$$np = 12, \quad \binom{n}{2}p^2 = 54 \xrightarrow{\text{match coefficients}} \frac{n(n-1)}{2}p^2 = 54 \xrightarrow{\text{solve}} n = 4, p = 3$$

Solution flow: each arrow applies one step.

(be-069)

Difficulty: Foundation / Marks: 4

Expand $(x + 1)^5 + (x - 1)^5$. Simplify your answer.

Worked solution.

Step 1: Expand both

$$(x+1)^5 = x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1$$

Notice odd powers of 1 cancel, even powers double.

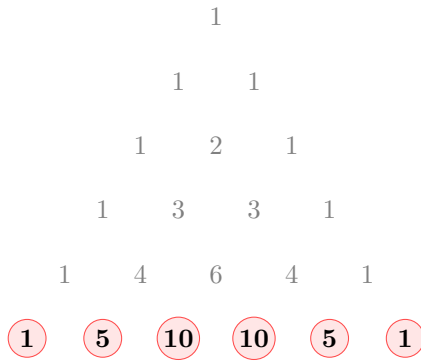
Step 2: Add

$$2x^5 + 20x^3 + 10x$$

Only even-powered terms survive.

Final answer: $2x^5 + 20x^3 + 10x$

Diagram.



Row 5 of Pascal's triangle (red) gives the binomial coefficients $\binom{5}{r}$; multiply each by the matching powers $a^{5-r}b^r$.

(be-070)

Difficulty: Foundation / *Marks: 3*

Find the coefficient of x^6 in $(1+x)^4(1+x)^5$. [Hint: $(1+x)^4(1+x)^5 = (1+x)^9$.]

Worked solution.

Step 1: Combine

$$(1+x)^9$$

Use the hint.

Step 2: Find coefficient

$$\binom{9}{6} = 84$$

Coefficient of x^6 is $\binom{9}{6}$.

Final answer: 84

Diagram.

$$(1+x)^4(1+x)^5 = (1+x)^9 \xrightarrow{\text{add the indices}} \binom{9}{6} = \binom{9}{3} \xrightarrow{\text{evaluate}} 84$$

Solution flow: each arrow applies one step.

Binomial Expansion 71 (be-071)

Difficulty: Challenge / Marks: 4

In the expansion of $(1+ax)^{10}$, the coefficient of x^2 is twice the coefficient of x . Find the value of a .

Worked solution.

Step 1: Apply the binomial theorem: $(1+ax)^{10} = \sum_{r=0}^{10} \binom{10}{r} (ax)^r$.

$$\text{Coeff of } x = 10a$$

$$\text{Coeff of } x^2 = 45a^2$$

The coefficient of x is $\binom{10}{1}a = 10a$. The coefficient of x^2 is $\binom{10}{2}a^2 = 45a^2$.

Step 2: Set up and solve the equation.

$$45a^2 = 2(10a)$$

$$45a^2 = 20a$$

$$45a^2 - 20a = 0$$

$$5a(9a - 4) = 0$$

$$a = \frac{4}{9}$$

The coefficient of x^2 is twice the coefficient of x .

Final answer: $a = \frac{4}{9}$
--

Common mistake. $a = 0$ is a trivial solution (all coefficients zero) so we discard it.

Diagram.

$$\binom{10}{2}a^2 = 2\binom{10}{1}a \xrightarrow{\text{set up the condition}} 45a^2 = 20a \xrightarrow{\text{solve } (a \neq 0)} a = \frac{4}{9}$$

Solution flow: each arrow applies one step.

Binomial Expansion 72 (be-072)

Difficulty: Challenge / Marks: 4

Find the coefficient of x^4 in the expansion of $(1+x)^5(1-x)^5$.

Worked solution.

Step 1: Simplify the product first using the difference of two squares.

$$(1+x)^5(1-x)^5 = (1-x^2)^5$$

$(1+x)^5(1-x)^5 = [(1+x)(1-x)]^5 = (1-x^2)^5$. This is much simpler than expanding each separately.

Step 2: Apply the binomial theorem: $(1+u)^n = \sum \binom{n}{r} u^r$ with $u = -x^2$.

$$\begin{aligned} \text{Coeff of } x^4 &= \binom{5}{2}(-1)^2 \\ &= 10 \end{aligned}$$

We need the coefficient of x^4 , which corresponds to $(-x^2)^2$, i.e. $r = 2$.

Final answer: 10

Diagram.

$$(1+x)^5(1-x)^5 = (1-x^2)^5 \xrightarrow{\text{difference of squares}} x^4 \text{ term : } \binom{5}{2}(-1)^2 \xrightarrow{\text{term in } x^4} 10$$

Solution flow: each arrow applies one step.

Binomial Expansion 73 (be-073)

Difficulty: Challenge / Marks: 5

The first three terms of $(1+px)^n$ are $1+20x+160x^2$. Find p and n .

Worked solution.

Step 1: Apply the binomial theorem: $(1+px)^n = 1 + np x + \frac{n(n-1)}{2} p^2 x^2 + \dots$

$$\begin{aligned} np &= 20 \quad \dots (1) \\ \frac{n(n-1)}{2} p^2 &= 160 \quad \dots (2) \end{aligned}$$

Match coefficients of x and x^2 to form simultaneous equations.

Step 2: Solve the simultaneous equations.

$$\begin{aligned} \frac{n(n-1)}{2} \cdot \frac{400}{n^2} &= 160 \\ \frac{200(n-1)}{n} &= 160 \\ 200n - 200 &= 160n \\ 40n &= 200 \\ n &= 5, \quad p = 4 \end{aligned}$$

From (1): $p = \frac{20}{n}$. Substitute into (2).

Final answer: $n = 5, p = 4$

Diagram.

$$np = 20, \binom{n}{2}p^2 = 160 \xrightarrow{\text{match coefficients}} \frac{n(n-1)}{2}p^2 = 160 \xrightarrow{\text{solve}} n = 5, p = 4$$

Solution flow: each arrow applies one step.

Binomial Expansion 74 (be-074)

Difficulty: Challenge / Marks: 5

Find the term independent of x in $\left(2x^2 + \frac{3}{x}\right)^6$.

Worked solution.

Step 1: Write the general term using the binomial theorem: $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$.

$$T_{r+1} = \binom{6}{r} (2x^2)^{6-r} \left(\frac{3}{x}\right)^r = \binom{6}{r} 2^{6-r} \cdot 3^r \cdot x^{12-2r-r} = \binom{6}{r} 2^{6-r} 3^r x^{12-3r}$$

Here $a = 2x^2$, $b = \frac{3}{x}$, $n = 6$. Combine powers of x .

Step 2: Set the power of x to zero.

$$12 - 3r = 0 \quad \Rightarrow \quad r = 4$$

Step 3: Substitute $r = 4$.

$$\begin{aligned} T_5 &= \binom{6}{4} \cdot 2^2 \cdot 3^4 \\ &= 15 \times 4 \times 81 \\ &= 4860 \end{aligned}$$

Final answer: 4860

Diagram.

$$T_{r+1} = \binom{6}{r} (2x^2)^{6-r} \left(\frac{3}{x}\right)^r$$

choose r power of 1st term power of 2nd term

General term of $\left(2x^2 + \frac{3}{x}\right)^6$. Set the power of x to the required value to find r , then evaluate.

Binomial Expansion 75 (be-075)

Difficulty: Challenge / Marks: 5

Given that $(1+kx)^8 = 1 + 12x + ax^2 + bx^3 + \dots$, find the values of k , a and b .

Worked solution.

Step 1: Apply the binomial theorem: $(1+kx)^8 = 1 + 8kx + \binom{8}{2}k^2x^2 + \binom{8}{3}k^3x^3 + \dots$

$$8k = 12 \quad \Rightarrow \quad k = \frac{3}{2}$$

Match the coefficient of x to find k .

Step 2: Find a and b .

$$a = \binom{8}{2} \left(\frac{3}{2}\right)^2 = 28 \times \frac{9}{4} = 63$$

$$b = \binom{8}{3} \left(\frac{3}{2}\right)^3 = 56 \times \frac{27}{8} = 189$$

Substitute $k = \frac{3}{2}$ into the coefficients of x^2 and x^3 .

Final answer: $k = \frac{3}{2}, a = 63, b = 189$

Diagram.

$$8k = 12 \Rightarrow k = \frac{3}{2} \xrightarrow{\text{find } k} a = \binom{8}{2} k^2 = 63 \xrightarrow{\text{find } a, b} b = \binom{8}{3} k^3 = 189$$

Solution flow: each arrow applies one step.

Binomial Expansion 76 (be-076)

Difficulty: Challenge / Marks: 5

Show that $(1 + \sqrt{2})^4 + (1 - \sqrt{2})^4 = 34$.

Worked solution.

Step 1: Expand both using the binomial theorem: $(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.

$$\begin{aligned} (1 + \sqrt{2})^4 &= 1 + 4\sqrt{2} + 6(2) + 4(2\sqrt{2}) + 4 \\ &= 1 + 4\sqrt{2} + 12 + 8\sqrt{2} + 4 = 17 + 12\sqrt{2} \\ (1 - \sqrt{2})^4 &= 1 - 4\sqrt{2} + 12 - 8\sqrt{2} + 4 = 17 - 12\sqrt{2} \end{aligned}$$

When we add the two expansions, all odd-powered terms in $\sqrt{2}$ cancel.

Step 2: Add the two expansions.

$$(17 + 12\sqrt{2}) + (17 - 12\sqrt{2}) = 34 \quad \checkmark$$

The surd terms cancel.

Final answer: 34 (shown)

Diagram.

$$(1 + \sqrt{2})^4 = 17 + 12\sqrt{2} \xrightarrow{\text{expand each}} (1 - \sqrt{2})^4 = 17 - 12\sqrt{2} \xrightarrow{\text{add (surds cancel)}} \text{sum} = 34$$

Solution flow: each arrow applies one step.

Binomial Expansion 77 (be-077)

Difficulty: Challenge / Marks: 6

The coefficient of x^3 in $(2+x)^5(1-x)^3$ is to be found. Expand each bracket up to x^3 and multiply.

Worked solution.

Step 1: Expand $(2+x)^5$ up to x^3 using $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$.

$$(2+x)^5 = 32 + 80x + 80x^2 + 40x^3 + \dots$$

Step 2: Expand $(1-x)^3$ fully.

$$(1-x)^3 = 1 - 3x + 3x^2 - x^3$$

Step 3: Collect all products that give x^3 .

$$\begin{aligned} & 32(-1) + 80(3) + 80(-3) + 40(1) \\ &= -32 + 240 - 240 + 40 = 8 \end{aligned}$$

The x^3 terms come from: $32 \times (-x^3)$, $80x \times 3x^2$, $80x^2 \times (-3x)$, $40x^3 \times 1$.

Final answer: 8

Diagram.

$$(2+x)^5 = 32 + 80x + 80x^2 + 40x^3 + \dots \xrightarrow{\text{expand each}} (1-x)^3 = 1 - 3x + 3x^2 - x^3 \xrightarrow{\text{collect } x^3} \text{coeff of } x^3 = 8$$

Solution flow: each arrow applies one step.

Binomial Expansion 78 (be-078)

Difficulty: Challenge / Marks: 3

Find the value of $\sum_{r=0}^6 \binom{6}{r}$.

Worked solution.

Use the binomial theorem: $(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$.

$$\sum_{r=0}^6 \binom{6}{r} = (1+1)^6 = 2^6 = 64$$

Setting $x = 1$ gives $\sum_{r=0}^n \binom{n}{r} = 2^n$. This is because each element is either included or excluded from a subset.

Final answer: 64

Diagram.

$$\sum_{r=0}^6 \binom{6}{r} \xrightarrow{\text{put } x=1 \text{ in } (1+x)^6} = (1+1)^6 \xrightarrow{\text{evaluate}} = 2^6 = 64$$

Solution flow: each arrow applies one step.

Binomial Expansion 79 (be-079)

Difficulty: Challenge / Marks: 4

In the expansion of $(1+x)^n$, the coefficients of x^4 and x^5 are equal. Find n .

Worked solution.

Apply the binomial theorem. The coefficients are $\binom{n}{4}$ and $\binom{n}{5}$.

$$\begin{aligned}\binom{n}{4} &= \binom{n}{5} \\ \frac{n!}{4!(n-4)!} &= \frac{n!}{5!(n-5)!} \\ \frac{1}{(n-4)!} \cdot \frac{5!}{4!} &= \frac{1}{(n-5)!} \\ 5 &= n-4 \\ n &= 9\end{aligned}$$

Set them equal and use the formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

Final answer: $n = 9$

Diagram.

$$\binom{n}{4} = \binom{n}{5} \xrightarrow{\text{equal coefficients}} 4 + 5 = n \xrightarrow{\text{symmetry of } \binom{n}{r}} n = 9$$

Solution flow: each arrow applies one step.

Binomial Expansion 80 (be-080)

Difficulty: Challenge / Marks: 4

Find the coefficient of x^2 in $(1+2x)^5 - (1-2x)^5$.

Worked solution.

Step 1: Apply the binomial theorem to each expansion up to x^2 .

$$\begin{aligned}(1+2x)^5 &= 1 + 10x + 40x^2 + \dots \\ (1-2x)^5 &= 1 - 10x + 40x^2 - \dots \\ \text{Difference: } &20x + 0x^2 + \dots\end{aligned}$$

When subtracting $(1+2x)^5 - (1-2x)^5$, the even-powered terms cancel and odd-powered terms double.

Step 2: The coefficient of x^2 is zero.

$$\text{Coefficient of } x^2 = 40 - 40 = 0$$

The x^2 terms are identical in both expansions (both $+40x^2$), so they cancel when subtracted.

Final answer: 0

Diagram.

$$(1 + 2x)^5 - (1 - 2x)^5 \xrightarrow{\text{subtract}} \text{even-power terms cancel} \xrightarrow{x^2 \text{ is an even power}} x^2 \text{ coeff} = 0$$

Solution flow: each arrow applies one step.

Binomial Expansion 81 (be-081)

Difficulty: Challenge / Marks: 5

The constant term in the expansion of $\left(x^2 - \frac{k}{x}\right)^9$ is 672. Find the possible values of k .

Worked solution.

Step 1: Write the general term using the binomial theorem.

$$T_{r+1} = \binom{9}{r} (x^2)^{9-r} \left(-\frac{k}{x}\right)^r = \binom{9}{r} (-k)^r x^{18-2r-r} = \binom{9}{r} (-k)^r x^{18-3r}$$

Here $a = x^2$, $b = -\frac{k}{x}$, $n = 9$.

Step 2: Set the power of x to zero for the constant term.

$$18 - 3r = 0 \quad \Rightarrow \quad r = 6$$

Step 3: Substitute $r = 6$ and set equal to 672.

$$\begin{aligned} \binom{9}{6} (-k)^6 &= 672 \\ 84k^6 &= 672 \\ k^6 &= 8 \\ k &= \sqrt[6]{8} = \sqrt{2} \end{aligned}$$

Final answer: $k = \sqrt{2}$ or $k = -\sqrt{2}$

Common mistake. Since $(-k)^6 = k^6$, both positive and negative values of k give the same constant term.

Diagram.

$$T_{r+1} = \binom{9}{r} (x^2)^{9-r} \left(-\frac{k}{x}\right)^r \xrightarrow{\text{general term}} 18 - 3r = 0 \Rightarrow r = 6 \xrightarrow{\text{constant term}} \binom{9}{6} k^6 = 672 \Rightarrow k = \pm\sqrt{2}$$

Solution flow: each arrow applies one step.

Binomial Expansion 82 (be-082)

Difficulty: Challenge / Marks: 4

Use the binomial expansion to show that $(1+x)^n + (1-x)^n = 2 \left[1 + \binom{n}{2}x^2 + \binom{n}{4}x^4 + \dots \right]$.

Worked solution.

Step 1: Expand both using the binomial theorem: $(1+x)^n = \sum_{r=0}^n \binom{n}{r}x^r$.

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots$$

$$(1-x)^n = \binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 - \binom{n}{3}x^3 + \dots$$

In $(1-x)^n$, the sign of each term alternates: $\binom{n}{r}(-x)^r = (-1)^r \binom{n}{r}x^r$.

Step 2: Add the two expansions.

$$(1+x)^n + (1-x)^n = 2 \left[1 + \binom{n}{2}x^2 + \binom{n}{4}x^4 + \dots \right] \quad \checkmark$$

The odd-powered terms have opposite signs and cancel. The even-powered terms are identical and double.

Final answer: Shown

Diagram.

$$(1+x)^n = \sum \binom{n}{r}x^r \xrightarrow{\text{expand both}} (1-x)^n = \sum \binom{n}{r}(-x)^r \xrightarrow{\text{add: odd powers cancel}} 2 \left[1 + \binom{n}{2}x^2 + \dots \right]$$

Solution flow: each arrow applies one step.

Binomial Expansion 83 (be-083)

Difficulty: Challenge / Marks: 6

Find the coefficient of x^6 in $(1+x+x^2)^5$.

Worked solution.

Step 1: Write $(1+x+x^2)^5 = [(1+x) + x^2]^5$ and apply the binomial theorem.

$$= \sum_{r=0}^5 \binom{5}{r} (1+x)^{5-r} x^{2r}$$

Let $a = 1+x$, $b = x^2$. Then $(a+b)^5 = \sum_{r=0}^5 \binom{5}{r} (1+x)^{5-r} (x^2)^r$.

Step 2: For x^6 , we need $2r+k=6$ where k is the power from $(1+x)^{5-r}$.

$r=0$: $k=6$, but $(1+x)^5$ max power is 5. Skip.

$r=1$: $k=4$, $\binom{5}{1} \binom{4}{4} = 5$

$r=2$: $k=2$, $\binom{5}{2} \binom{3}{2} = 10 \times 3 = 30$

$r=3$: $k=0$, $\binom{5}{3} \binom{2}{0} = 10$

Check each valid r :

Step 3: Sum the contributions.

$$5 + 30 + 10 = 45$$

Final answer: 45

Diagram.

$$(1 + x + x^2)^5 \xrightarrow{\text{multinomial}} \text{count ways to make } x^6 \xrightarrow{\text{coefficient}} 45$$

Solution flow: each arrow applies one step.

Binomial Expansion 84 (be-084)

Difficulty: Challenge / Marks: 5

The ratio of the coefficient of x^3 to the coefficient of x^4 in $(1 + 2x)^n$ is $2 : 3$. Find n .

Worked solution.

Step 1: Apply the binomial theorem: the general term is $\binom{n}{r}(2x)^r$.

$$\text{Coeff of } x^3 = 8 \binom{n}{3}$$

$$\text{Coeff of } x^4 = 16 \binom{n}{4}$$

The coefficients of x^3 and x^4 are $\binom{n}{3} \cdot 8$ and $\binom{n}{4} \cdot 16$.

Step 2: Set up the ratio and solve.

$$\begin{aligned} \frac{8 \binom{n}{3}}{16 \binom{n}{4}} &= \frac{2}{3} \\ \frac{8}{16} \cdot \frac{4}{n-3} &= \frac{2}{3} \\ \frac{2}{n-3} &= \frac{2}{3} \\ n-3 &= 3 \\ n &= 6 \end{aligned}$$

Use $\binom{n}{4} = \frac{n-3}{4} \binom{n}{3}$.

Final answer: $n = 6$

Diagram.

$$\frac{\binom{n}{3} 2^3}{\binom{n}{4} 2^4} = \frac{2}{3} \xrightarrow{\text{ratio of coefficients}} \frac{4}{2(n-3)} = \frac{2}{3} \xrightarrow{\text{solve}} n = 6$$

Solution flow: each arrow applies one step.

Binomial Expansion 85 (be-085)

Difficulty: Challenge / Marks: 4

Use the expansion of $(1+x)^{10}$ with a suitable value of x to find $\sum_{r=0}^{10} (-1)^r \binom{10}{r} 3^r$.

Worked solution.

Recognise the sum as a binomial expansion.

$$\begin{aligned}\sum_{r=0}^{10} (-1)^r \binom{10}{r} 3^r &= \sum_{r=0}^{10} \binom{10}{r} (-3)^r \\ &= (1 + (-3))^{10} \\ &= (-2)^{10} = 1024\end{aligned}$$

By the binomial theorem, $(1+x)^{10} = \sum_{r=0}^{10} \binom{10}{r} x^r$. The given sum has $(-1)^r \cdot 3^r = (-3)^r$, so set $x = -3$.

Final answer: 1024

Diagram.

$$\sum_{r=0}^{10} (-1)^r \binom{10}{r} 3^r \xrightarrow{\text{put } x = -3 \text{ in } (1+x)^{10}} (1-3)^{10} \xrightarrow{\text{evaluate}} (-2)^{10} = 1024$$

Solution flow: each arrow applies one step.
